

CHAPTER

13

Identical Particles

To study systems like multielectron atoms, we need to properly account for the fact that all fundamental particles like electrons and protons are **identical**. In classical physics, particles are not identical—we can always find a way to uniquely identify a particular particle. Even if we make two classical particles “the same” to the utmost level of precision, we can still find a way to identify the two particles without affecting their classical motion. For example, billiard balls behave identically, but can be identified by their numbers. In quantum mechanics, there is no way to identify two different electrons—they are **indistinguishable**. Two hydrogen atoms are identical no matter where they are in the universe. Researchers rely on this fact when they compare their experimental results on the spectra of hydrogen atoms in different laboratories. To account for the indistinguishability of fundamental particles, we introduce a new postulate in quantum mechanics, which leads to the Pauli exclusion principle that is responsible for the periodic table and all of chemistry. We apply this new postulate to the helium atom to learn how the indistinguishability of the two electrons in the atom affects the energies and the allowed states.

13.1 ■ TWO SPIN-1/2 PARTICLES

To start our discussion of identical particles, let’s return to the system of two spin-1/2 particles that we studied in Chapter 11. We found that we could describe the system using either of two bases:

$$\begin{array}{ll} |++\rangle, |+-\rangle, |-+\rangle, |--\rangle & \text{uncoupled basis } |s_1 s_2 m_1 m_2\rangle \\ |11\rangle, |10\rangle, |1,-1\rangle, |00\rangle & \text{coupled basis } |SM_S\rangle. \end{array} \quad (13.1)$$

The coupled basis is preferred when the two particles or systems interact, such as in the hyperfine interaction or the spin-orbit interaction, because the Hamiltonian is diagonal in that basis. In general, our choice of basis depends on the problem at hand, but that choice is one we make based solely on convenience in finding the energy eigenvalues. If we choose the other basis, we still find the correct eigenvalues—it just takes a little more work. However, if the two particles are *identical*, then that *freedom of choice of basis is no longer available*. Let’s see why.

Consider the ket $|+-\rangle$ in the uncoupled basis that represents a quantum state in which particle 1 has spin up and particle 2 has spin down. If the two particles in question are a proton and an electron, then this representation is clear and unambiguous. However, if the two particles are electrons, then this representation is more problematic. How can we possibly know that the particle that we measure to have spin up is electron 1 and not electron 2? *We cannot*. In quantum mechanics there is no way to distinguish the two particles, and we cannot tell which particle has spin up and which has spin down.

Or as Dr. Seuss said, we cannot know “*Whether this one was that one . . . or that one was this one Or which one was what one . . . or what one was who.*” There is no experiment we can perform on the system of two electrons that would distinguish the state $|+-\rangle$ from the state $| - + \rangle$. This leads us to conclude that the uncoupled basis is inappropriate for representing this system of two identical spin-1/2 particles. So how do we mathematically represent the state with one particle having spin up and one particle having spin down?

The best way would be to start over and abandon the attempt at labeling the quantum numbers of individual particles and instead specify how many particles have particular sets of quantum numbers. This approach is the basis of more advanced treatments, but is too much of a change for us to make at this stage. So we adapt our labeling scheme to this new problem. A reasonable guess for representing the state with one particle having spin up and one particle having spin down would be to use a superposition of the two states $|+-\rangle$ and $| - + \rangle$ in a way that does not favor one over the other. From Chapter 11 on the addition of angular momenta, we know that such superpositions already exist in the coupled basis representation:

$$\begin{aligned} |10\rangle &= \frac{1}{\sqrt{2}}[|+-\rangle + |-+\rangle] \\ |00\rangle &= \frac{1}{\sqrt{2}}[|+-\rangle - |-+\rangle]. \end{aligned} \quad (13.2)$$

These two states differ only by the minus sign coefficient. Let's see what the importance of that minus sign is, and in doing so, learn why the coupled basis is appropriate for describing systems of identical particles.

Imagine that, unbeknownst to us, someone *exchanged* the two identical particles in the system, so that what we originally thought was particle 1 is now particle 2 and vice versa. The two states in Eq. (13.2) would then become

$$\begin{aligned} \frac{1}{\sqrt{2}}[|+-\rangle + |-+\rangle] &\xrightarrow{\text{exchange}} \frac{1}{\sqrt{2}}[|-+\rangle + |+-\rangle] = \frac{1}{\sqrt{2}}[|+-\rangle + |-+\rangle] \\ \frac{1}{\sqrt{2}}[|+-\rangle - |-+\rangle] &\xrightarrow{\text{exchange}} \frac{1}{\sqrt{2}}[|-+\rangle - |+-\rangle] = -\frac{1}{\sqrt{2}}[|+-\rangle - |-+\rangle]. \end{aligned} \quad (13.3)$$

The first state is unchanged and the second state acquires a minus sign, which is an overall phase shift of 180° . An overall phase shift causes no measurable change (Problem 1.3), so the physical states are unchanged by this exchange operation. We denote the exchange operation by the **exchange operator** P_{12} , whose action on uncoupled basis states is

$$P_{12}|s_1s_2m_1m_2\rangle = |s_2s_1m_2m_1\rangle. \quad (13.4)$$

In terms of the coupled basis representation, the action of the exchange operator P_{12} on the two states in Eq. (13.2) is

$$\begin{aligned} P_{12}|10\rangle &= +|10\rangle \\ P_{12}|00\rangle &= -|00\rangle. \end{aligned} \quad (13.5)$$

These results tell us that the coupled basis states $|10\rangle$ and $|00\rangle$ are eigenstates of the exchange operator with eigenvalues $+1$ and -1 , respectively. We call these states *symmetric* ($|10\rangle$) and *antisymmetric* ($|00\rangle$) states, by which we mean that they are symmetric and antisymmetric with respect to the exchange of the two particles, as opposed to symmetric and antisymmetric with respect to space or

something else. Note that the other two coupled basis states $|11\rangle$ and $|1,-1\rangle$ are both symmetric with respect to exchange, (i.e., have eigenvalues of $+1$):

$$\begin{aligned} P_{12}|11\rangle &= P_{12}|++\rangle = |++\rangle = +|11\rangle \\ P_{12}|1,-1\rangle &= P_{12}|--\rangle = |--\rangle = +|1,-1\rangle. \end{aligned} \quad (13.6)$$

Now consider a measurement upon an eigenstate of the exchange operator, which we denote by $|\psi_{\pm}\rangle$, where the $\pm$ indicates the eigenvalue. The probability of recording some final result is

$$\mathcal{P}_{\psi_{\pm} \rightarrow \psi_f} = \left| \langle \psi_f | \psi_{\pm} \rangle \right|^2. \quad (13.7)$$

If we exchange the two particles before the measurement, then the probability is

$$\begin{aligned} \mathcal{P}_{P_{12}\psi_{\pm} \rightarrow \psi_f} &= \left| \langle \psi_f | \{P_{12}|\psi_{\pm}\} \rangle \right|^2 = \left| \langle \psi_f | P_{12}|\psi_{\pm} \rangle \right|^2 \\ &= \left| \langle \psi_f | (\pm 1)|\psi_{\pm} \rangle \right|^2 = \left| \langle \psi_f | \psi_{\pm} \rangle \right|^2 \\ &= \mathcal{P}_{\psi_{\pm} \rightarrow \psi_f}. \end{aligned} \quad (13.8)$$

Hence, the calculated probability for a measurement made upon a state $|\psi_{\pm}\rangle$ is not changed if the particles are exchanged. This agrees with our statement above that experiments cannot distinguish between systems with the identical particles exchanged. Thus the coupled basis superpositions are promising representations of the physical system of identical particles.

At this point, both coupled states $|10\rangle$ and $|00\rangle$ plausibly represent the physical state of a system of two identical spin-1/2 particles with one particle having spin up and one having spin down. However, we know from Chapter 11 on angular momentum addition that these two states have an important difference. The state $|10\rangle$ has total spin angular momentum $S = 1$, while the state $|00\rangle$ has total spin angular momentum $S = 0$. Thus, they are clearly not describing the same system. So how do we know which of these two states to use to describe our system of two electrons with one having spin up and the other having spin down?

Nature chooses for us. You probably already know about the **Pauli exclusion principle** that forbids having two electrons in the same quantum state. The Pauli exclusion principle is a specific example of a broader quantum mechanical principle that we call the **symmetrization postulate**.

The symmetrization postulate stipulates that a system of identical particles is required to have a quantum state vector that is either symmetric or antisymmetric with respect to exchange of any pair of particles. Nature has sorted particles into two classes depending on whether they obey the symmetric or antisymmetric version of this principle. Particles that are required to have symmetric states are called **bosons** and particles that are required to have antisymmetric states are called **fermions**. Furthermore, this symmetry property is correlated with the spin angular momentum of the particles comprising the system. Bosons are particles with integer spin (0, 1, 2, ...) and are required to have symmetric quantum states. Fermions are particles with half-integer spin (1/2, 3/2, 5/2, ...) and are required to have antisymmetric quantum states. This connection between the spins of particles and their exchange symmetry can be proved using relativistic quantum mechanics, so we take it as a postulate. It has been confirmed in many experiments. Because the exchange symmetry determines the statistical behavior of these particles, this concept is often called the **spin-statistics theorem**.

Electrons have spin 1/2, so they are fermions, as are protons and neutrons. Photons and mesons are examples of bosons. Composite particles, like atoms, are fermions or bosons, depending on the total

spin of the system. For example, hydrogen has one electron and one proton—two spin-1/2 fermions—so it has integer total spin (1 or 0) and is a boson. Deuterium has one electron and a nucleus (the deuteron) comprising one proton and one neutron, so it has half-integer spin and is a fermion. Thus, different isotopes of the same atom can behave differently when one considers the collective behavior of atoms. For example, samples of liquid ^3He (fermion) and liquid ^4He (boson) behave quite differently at very low temperatures.

To summarize, the symmetrization postulate tells us that the quantum state vector of a system of two (or more) identical particles must be either symmetric or antisymmetric with respect to exchange of the two (or any two) particles. All particles are divided into either fermions or bosons:

Bosons (integer spin: 0, 1, 2, ...) must have symmetric states.

Fermions (half-integer spin: 1/2, 3/2, 5/2, ...) must have antisymmetric states.

For the system of two identical spin-1/2 particles we discussed above, you would then conclude that the two particles must be in the antisymmetric state $|00\rangle$, and the symmetric states $|11\rangle$, $|10\rangle$, and $|1, -1\rangle$ are not allowed. However, we cannot yet reach that conclusion because the symmetrization postulate applies to the complete state vector, and we have not yet included the spatial part of the state vector.

We assume that we can separate the spin and spatial aspects of the state vectors. This is not always possible, but it works for all the systems we study. The complete quantum state vector then has the form

$$|\psi\rangle = |\psi_{\text{spatial}}\rangle |\psi_{\text{spin}}\rangle. \quad (13.9)$$

The symmetrization postulate must be applied to this complete quantum state. For *bosons*, the complete state vector must be *symmetric* under exchange of particles. Thus, the spatial part must be symmetric if the spin part is symmetric, or the spatial part must be antisymmetric if the spin part is antisymmetric:

$$\begin{aligned} |\psi_{\text{boson}}^{SS}\rangle &= |\psi_{\text{spatial}}^S\rangle |\psi_{\text{spin}}^S\rangle \\ |\psi_{\text{boson}}^{AA}\rangle &= |\psi_{\text{spatial}}^A\rangle |\psi_{\text{spin}}^A\rangle. \end{aligned} \quad (13.10)$$

The two parts *must* have the same exchange symmetry, or else the full eigenstate would not be symmetric. For *fermions*, the complete state vector must be *antisymmetric* under exchange. Thus, the spatial part must be symmetric if the spin part is antisymmetric, or the spatial part must be antisymmetric if the spin part is symmetric:

$$\begin{aligned} |\psi_{\text{fermion}}^{SA}\rangle &= |\psi_{\text{spatial}}^S\rangle |\psi_{\text{spin}}^A\rangle \\ |\psi_{\text{fermion}}^{AS}\rangle &= |\psi_{\text{spatial}}^A\rangle |\psi_{\text{spin}}^S\rangle. \end{aligned} \quad (13.11)$$

The two parts *cannot* have the same exchange symmetry, or else the full eigenstate would not be antisymmetric. In Eqs. (13.10) and (13.11) we use superscripts to denote the exchange symmetry of the state vectors.

13.2 ■ TWO IDENTICAL PARTICLES IN ONE DIMENSION

We take the example of a system of two identical particles bound within a one-dimensional potential energy well to study the application of the symmetrization postulate to the complete state vector. Limiting the discussion to one dimension is sufficient to illustrate the most important ramifications of the symmetrization principle. Let's first look at the spatial part of the state vector. The Hamiltonian for a single particle in one dimension is

$$H_{\text{single}} = \frac{p^2}{2m} + V(x). \quad (13.12)$$

Assume we know the wave function solutions to the energy eigenvalue equation:

$$H_{\text{single}} \varphi_n(x) = E_n \varphi_n(x). \quad (13.13)$$

The Hamiltonian for two particles in this potential energy well is

$$H = \frac{p_1^2}{2m} + V(x_1) + \frac{p_2^2}{2m} + V(x_2), \quad (13.14)$$

where x_1 labels the position of particle 1 along the x -axis, and p_1 is the momentum of particle 1; x_2 and p_2 are the equivalent for particle 2. We assume for the moment that the two particles do not interact with each other and that the Hamiltonian has no spin dependence. The two-particle energy eigenvalue equation is

$$H\psi(x_1, x_2) = E\psi(x_1, x_2). \quad (13.15)$$

The wave function $\psi(x_1, x_2)$ of the system is a function of the two coordinates x_1 and x_2 locating the two particles. This **two-particle wave function** is a new concept, so a few comments about it are in order.

The complex square of the two-particle wave function yields the probability density

$$\mathcal{P}(x_1, x_2) = |\psi(x_1, x_2)|^2, \quad (13.16)$$

but because we have two particles, we must be clear what this density means. We interpret the probability density as a **two-particle probability density**, and so

$$|\psi(x_1, x_2)|^2 dx_1 dx_2 \quad (13.17)$$

is the probability of finding particle 1 at position x_1 within a volume dx_1 and finding particle 2 at position x_2 within a volume dx_2 . ("Volume" in this case is a length, but in a three-dimensional problem, it would be a true volume.) We normalize the system wave function by integrating the two-particle probability density over both coordinates:

$$\iint |\psi(x_1, x_2)|^2 dx_1 dx_2 = 1, \quad (13.18)$$

which means that the probability of finding both particles within the whole volume available to the system is unity.

In this noninteracting example, the two-particle Hamiltonian is the sum of two single-particle Hamiltonians, so the product function $\varphi_{n_a}(x_1)\varphi_{n_b}(x_2)$ describing the system with particle 1 in energy state n_a and particle 2 in energy state n_b satisfies the energy eigenvalue equation (13.15) with energy $E_{n_a} + E_{n_b}$. That would be the end of the story if the two particles were *distinguishable* (like an electron and a proton, for example), but for *indistinguishable* or *identical* particles we must find spatial eigenstates that are either symmetric or antisymmetric with respect to exchange of the two particles. As we did with the spin state vectors for the spin-1/2 states in the last section, we form the symmetric or antisymmetric superpositions

$$\begin{aligned} |\psi_{space}^S\rangle &\doteq \psi_{n_a n_b}^S(x_1, x_2) = N_S [\varphi_{n_a}(x_1)\varphi_{n_b}(x_2) + \varphi_{n_b}(x_1)\varphi_{n_a}(x_2)] \\ |\psi_{space}^A\rangle &\doteq \psi_{n_a n_b}^A(x_1, x_2) = N_A [\varphi_{n_a}(x_1)\varphi_{n_b}(x_2) - \varphi_{n_b}(x_1)\varphi_{n_a}(x_2)], \end{aligned} \quad (13.19)$$

where $N_{S,A}$ are the normalization constants. The wave function $\psi_{n_a n_b}^S(x_1, x_2)$ is symmetric with respect to exchange of the two particles, and the wave function $\psi_{n_a n_b}^A(x_1, x_2)$ is antisymmetric. Each of these solutions has the two-particle energy

$$E_{n_a n_b} = E_{n_a} + E_{n_b}. \quad (13.20)$$

13.2.1 ■ Two-Particle Ground State

The ground state of this system has both particles in the single-particle ground state, so $n_a = 1$ and $n_b = 1$ and the energy of the state is $E_{11} = 2E_1$. The symmetric two-particle ground-state wave function is

$$\psi_{11}^S(x_1, x_2) = \varphi_1(x_1)\varphi_1(x_2). \quad (13.21)$$

However, the antisymmetric two-particle wave function is identically equal to zero:

$$\psi_{11}^A(x_1, x_2) = N_A [\varphi_1(x_1)\varphi_1(x_2) - \varphi_1(x_2)\varphi_1(x_1)] = 0, \quad (13.22)$$

so there is no possibility of having an antisymmetric spatial wave function in the ground state, regardless of whether the system comprises bosons or fermions.

To properly apply the symmetrization postulate to the complete state vector of this system, we must include the spin in the state vector. Let's assume that the system is either composed of two spin-0 bosons, or two spin-1/2 fermions. For two spin-0 bosons, the total spin must be zero and the only possible system spin state is $|SM\rangle = |00\rangle$, which is equal to the uncoupled basis state $|s_1 s_2 m_1 m_2\rangle = |0000\rangle$. This spin state is symmetric under exchange of the two particles (Problem 13.2). The complete state vector for bosons must be symmetric, so the spatial wave function must always be symmetric in this spin-0 example. Hence, the ground state of the two spin-0 bosons in a one-dimensional system is

$$|\psi_{11}^{SS}\rangle \doteq \psi_{11}^S(x_1, x_2)|00\rangle = \varphi_1(x_1)\varphi_1(x_2)|00\rangle. \quad (13.23)$$

The notation in Eq. (13.23) is a mixture of wave function language and abstract ket notation, but it makes the space-spin distinction clear.

For two spin-1/2 fermions, the total spin is 0 or 1, with the coupled basis states

$$\left. \begin{aligned} |11\rangle &= |++\rangle \\ |10\rangle &= \frac{1}{\sqrt{2}}[|+-\rangle + |-+\rangle] \\ |1,-1\rangle &= |--\rangle \\ |00\rangle &= \frac{1}{\sqrt{2}}[|+-\rangle - |-+\rangle] \end{aligned} \right\} \begin{array}{l} \text{Symmetric Triplet states} \\ \text{Antisymmetric Singlet state} \end{array} \quad (13.24)$$

being the eigenstates of the exchange operator we need to construct the complete state vectors. For the ground state, the symmetric spin triplet states are excluded because the required antisymmetric spatial state is identically zero [Eq. (13.22)]. The ground state of two spin-1/2 fermions is therefore

$$|\psi_{11}^{SA}\rangle \doteq \psi_{11}^S(x_1, x_2)|00\rangle = \varphi_1(x_1)\varphi_1(x_2)|00\rangle. \quad (13.25)$$

This result exposes a problem with our notation. The spin state $|00\rangle$ in Eq. (13.23) is *not* the same as the spin state $|00\rangle$ in Eq. (13.25). For two spin-0 bosons, the state $|00\rangle$ is really $|s_1 = 0, s_2 = 0, S = 0, M = 0\rangle$ and is symmetric under particle exchange, whereas for two spin-1/2 fermions, the state $|00\rangle$ is $|s_1 = \frac{1}{2}, s_2 = \frac{1}{2}, S = 0, M = 0\rangle$ and is antisymmetric under particle exchange. We will continue with the notation $|SM\rangle$ for coupled basis states, but note this limitation.

13.2.2 ■ Two-Particle Excited State

The first excited state of the two-particle system has one particle in the single-particle ground state and one particle in the first single-particle excited state, so $n_a = 1$, $n_b = 2$ and the energy of the state is $E_{12} = E_1 + E_2$. In this case, both the symmetric and antisymmetric spatial wave functions in Eq. (13.19) are nonzero.

For the spin-0 boson case, the spatial wave function must be symmetric because there is only a symmetric spin state, so the state vector is

$$|\psi_{12}^{SS}\rangle \doteq \psi_{12}^S(x_1, x_2)|00\rangle = \frac{1}{\sqrt{2}}[\varphi_1(x_1)\varphi_2(x_2) + \varphi_2(x_1)\varphi_1(x_2)]|00\rangle. \quad (13.26)$$

For the spin-1/2 fermion case, the total state vector must be antisymmetric, so the symmetric spatial wave function must combine with the antisymmetric singlet spin state

$$|\psi_{12}^{SA}\rangle \doteq \psi_{12}^S(x_1, x_2)|00\rangle = \frac{1}{\sqrt{2}}[\varphi_1(x_1)\varphi_2(x_2) + \varphi_1(x_2)\varphi_2(x_1)]|00\rangle, \quad (13.27)$$

and the antisymmetric spatial wave function must combine with the symmetric triplet spin states

$$|\psi_{12}^{AS}\rangle \doteq \psi_{12}^A(x_1, x_2)|1M\rangle = \frac{1}{\sqrt{2}}[\varphi_1(x_1)\varphi_2(x_2) - \varphi_1(x_2)\varphi_2(x_1)]1M\rangle, \quad (13.28)$$

with $M = 1, 0, -1$. The first excited state of the fermion system is four-fold degenerate, while the boson state is nondegenerate.

If the two particles were distinguishable, then the first excited state would be two-fold degenerate (assuming no spin), with states $\psi_{12}(x_1, x_2) = \varphi_1(x_1)\varphi_2(x_2)$ and $\psi_{21}(x_1, x_2) = \varphi_2(x_1)\varphi_1(x_2)$. A schematic of the ground and first excited states for all three cases is shown in Fig. 13.1. For the spin-1/2 fermions in Fig. 13.1(c), we use arrows to indicate the spin combinations. The states $|11\rangle$ and $|1, -1\rangle$ have the two spins aligned, in which case the notation is clear. However, the states $|10\rangle$ and $|00\rangle$ are different superpositions of spin up and down states, so the notation is somewhat unclear. You must remember that $|10\rangle = \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle)$ and $|00\rangle = \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle)$.

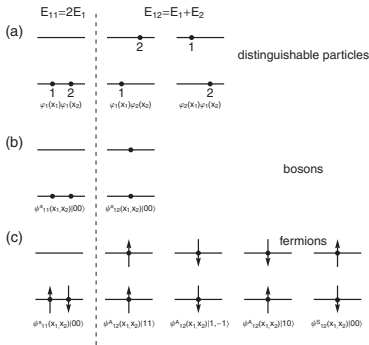


FIGURE 13.1 Schematic diagrams of ground and first excited states of two particles in a one-dimensional well for (a) distinguishable particles, (b) identical spin-0 bosons, and (c) identical spin-1/2 fermions.

13.2.3 Visualization of States

To visualize the spatial aspect of these states, assume the two particles are bound in an infinite square potential energy well. The one-particle energy eigenstate wave functions for the infinite square well are

$$|n\rangle \doteq \varphi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right). \quad (13.29)$$

The spatial wave function for the ground state is the same [Eq. (13.21)] for all three cases of distinguishable particles, identical bosons, and identical fermions. The two-particle probability density for the ground state is thus

$$\begin{aligned} \mathcal{P}(x_1, x_2) &= |\psi(x_1, x_2)|^2 \\ &= |\varphi_1(x_1)\varphi_1(x_2)|^2 \\ &= \frac{4}{L^2} \sin^2\left(\frac{\pi x_1}{L}\right) \sin^2\left(\frac{\pi x_2}{L}\right), \end{aligned} \quad (13.30)$$

as shown in Fig. 13.2. The system probability density is two-dimensional because there are two particles, each with a one-dimensional probability density.

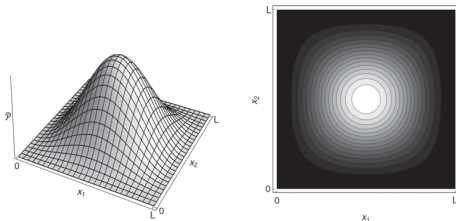


FIGURE 13.2 Two-particle spatial probability density for the ground state of a system of two particles in an infinite square well, displayed using height (left) or grayscale with contours (right). This probability density is the same for distinguishable particles, identical bosons, and identical fermions.

For the first excited state of the system with one particle in the single-particle ground state and one particle in the first single-particle excited state, the wave function does depend on the type of particle, as shown in Fig. 13.1. For the distinguishable particle case, there are two possible states, shown in Fig. 13.3(a) and (b). For the spin-0 boson case, there is only one possible state, which is the symmetric wave function $\psi_{12}^S(x_1, x_2)$ given in Eq. (13.26) with the probability density shown in Fig. 13.3(c). For the case of two spin-1/2 fermions, the excited state can either be in the symmetric [Fig. 13.3(c)] or antisymmetric [Fig. 13.3(d)] spatial wave function, depending on the spin state as given in Eqs. (13.27) and (13.28), respectively. For bosons and fermions, the *probability density* is symmetric with respect to particle exchange, which is evident in the symmetry about the diagonal line $x_1 = x_2$ in Figs. 13.3(c) and (d). The wave function $\psi_{12}^A(x_1, x_2)$ that underlies the probability density in Fig. 13.3(d), is antisymmetric about the line $x_1 = x_2$, but its square—the *probability density*—is symmetric. The probability density of the asymmetric spatial state $\psi_{12}^A(x_1, x_2)$ is identically zero along the line $x_1 = x_2$:

$$\psi_{12}^A(x_1, x_1) = \frac{1}{\sqrt{2}}[\varphi_1(x_1)\varphi_2(x_1) - \varphi_1(x_1)\varphi_2(x_1)] = 0, \quad (13.31)$$

illustrating that two fermions in a symmetric spin state cannot be in the same location. This is the Pauli exclusion principle that two electrons with the same spin orientation ($|11\rangle = |++\rangle$ or $|1, -1\rangle = |--\rangle$ states) cannot be in the same spatial state. The symmetrization postulate tells us that this also applies to two electrons with opposite spin but combined in a symmetric manner ($|10\rangle$ state). The spatial probability density shown in Fig. 13.3(d) illustrates the idea that fermions in a symmetric spin state appear to “repel” each other. In contrast, two fermions with opposite spins combined in an antisymmetric manner ($|00\rangle$ state) to make a spin-singlet state have a symmetric spatial wave function and the probability density shown in Fig. 13.3(c), the same as two bosons. In this case, the probability density is peaked along the line $x_1 = x_2$, illustrating that two bosons or two spin-singlet fermions have a tendency to “attract” each other.

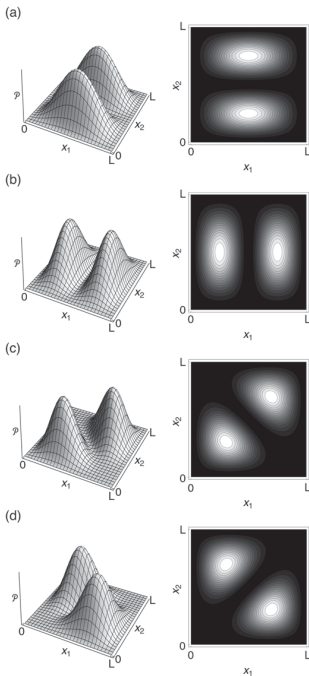


FIGURE 13.3 Two particle probability densities for the first excited state of a system of two particles in an infinite square well. (a, b) Distinguishable particles, (c) Symmetric spatial state for spin-0 bosons or spin-singlet fermions, (d) Anti-symmetric spatial state for spin-triplet fermions.

13.2.4 ■ Exchange Interaction

This apparent spatial attraction between bosons and spin-singlet fermions and repulsion between spin-triplet fermions does not reflect any potential energy of interaction between the two particles—we have assumed that the particles *do not interact* in this simple model. Rather, this *apparent interaction* is a consequence of the symmetrization requirement imposed on the wave functions. This effect is called the **exchange force** or the **exchange interaction**. One way to quantify the exchange interaction and demonstrate the difference between particles in symmetric and antisymmetric spatial states is to calculate the expectation value of the square of the separation of the two particles

$$\begin{aligned}\langle (x_1 - x_2)^2 \rangle &= \langle x_1^2 - 2x_1x_2 + x_2^2 \rangle \\ &= \langle x_1^2 \rangle + \langle x_2^2 \rangle - 2\langle x_1x_2 \rangle.\end{aligned}\quad (13.32)$$

We'll leave the bulk of this calculation to you, but let's demonstrate how to calculate one of these two-particle expectation values.

Consider the expectation value $\langle x_1^2 \rangle$ in the fermionic state $|\psi_{12}^{AS}\rangle = |\psi_{12}^A\rangle|1M\rangle$, where $|\psi_{12}^A\rangle \doteq \psi_{12}^A(x_1, x_2)$ is the spatial part. Using Dirac bra-ket notation to begin, we have

$$\begin{aligned}\langle x_1^2 \rangle &= \langle \psi_{12}^{AS} | x_1^2 | \psi_{12}^{AS} \rangle \\ &= \langle \psi_{12}^A | \langle 1M | x_1^2 | \psi_{12}^A \rangle | 1M \rangle.\end{aligned}\quad (13.33)$$

We separate the space and spin parts of the matrix element and recall that the position x_1 does not act on the spin states, so

$$\langle x_1^2 \rangle = \langle \psi_{12}^A | x_1^2 | \psi_{12}^A \rangle \langle 1M | 1M \rangle. \quad (13.34)$$

The spin state projection is unity: $\langle 1M | 1M \rangle = 1$. Let's keep the spatial matrix element in Dirac notation by using the notation $|n\rangle_1 \doteq \varphi_n(x_1)$, such that $|1\rangle_1 \doteq \varphi_1(x_1)$ and $|1\rangle_2 \doteq \varphi_1(x_2)$. The Dirac ket representation of the two-particle spatial state [Eq. (13.28)] in terms of the single-particle spatial states $[|\psi_{12}^A\rangle = \frac{1}{\sqrt{2}}(|1\rangle_1|2\rangle_2 - |2\rangle_1|1\rangle_2)]$ yields

$$\begin{aligned}\langle x_1^2 \rangle &= \frac{1}{\sqrt{2}} \langle {}_1\langle 1 | {}_2\langle 2 | - {}_2\langle 1 | {}_1\langle 2 | \rangle (x_1^2) \frac{1}{\sqrt{2}} (|1\rangle_1|2\rangle_2 - |2\rangle_1|1\rangle_2) \\ &= \frac{1}{2} \{ \langle {}_1\langle 1 | x_1^2 | {}_1\rangle_1 \rangle \langle {}_2\langle 2 | {}_2\rangle_2 \rangle - \langle {}_1\langle 1 | x_1^2 | {}_2\rangle_1 \rangle \langle {}_2\langle 2 | {}_1\rangle_2 \rangle \\ &\quad - \langle {}_1\langle 2 | x_1^2 | {}_1\rangle_1 \rangle \langle {}_2\langle 1 | {}_2\rangle_2 \rangle + \langle {}_1\langle 2 | x_1^2 | {}_2\rangle_1 \rangle \langle {}_2\langle 1 | {}_1\rangle_2 \rangle \},\end{aligned}\quad (13.35)$$

where we have isolated the separate matrix elements and projections for particles 1 and particle 2. Invoking the orthonormality of the single-particle eigenstates yields

$$\langle x_1^2 \rangle = \frac{1}{2} (\langle {}_1\langle 1 | x_1^2 | {}_1\rangle_1 \rangle + \langle {}_1\langle 2 | x_1^2 | {}_2\rangle_1 \rangle). \quad (13.36)$$

Thus we are left with calculating two single-particle expectation values. The subscript label indicating the particle number is irrelevant for that calculation, leaving

$$\langle x_1^2 \rangle = \frac{1}{2} (\langle {}_1\langle 1 | x^2 | {}_1\rangle_1 \rangle + \langle {}_2\langle 2 | x^2 | {}_2\rangle_2 \rangle). \quad (13.37)$$

To calculate the single-particle expectation values for the particle in the infinite square well, we must use an integral in the position representation:

$$\langle n | x^2 | n \rangle = \int_0^L \varphi_n^*(x) x^2 \varphi_n(x) dx. \quad (13.38)$$

Following this example, you can calculate the expectation value of the interparticle spacing in Eq. (13.32). The difference between the results for different spatial states resides in the cross-term $\langle x_1 x_2 \rangle$. The final result for the state with one particle in the $n = 1$ state and one particle in the $n = 2$ state of the infinite square well is (Problem 13.6)

$$\begin{aligned} \sqrt{\langle (x_1 - x_2)^2 \rangle_S} &= 0.20L \\ \sqrt{\langle (x_1 - x_2)^2 \rangle_D} &= 0.32L \\ \sqrt{\langle (x_1 - x_2)^2 \rangle_A} &= 0.41L \end{aligned} \quad (13.39)$$

for the three cases of distinguishable particles (D), identical particles in symmetric spatial states (S) (bosons or spin-singlet fermions), and identical particles in antisymmetric spatial states (A) (spin-triplet fermions). These results indicate that particles in symmetric spatial states (typically bosons) are closer to each other, and particles in antisymmetric spatial states (typically fermions) are farther apart from each other, compared to the distinguishable particle case.

The relation between the symmetry/antisymmetry of the spatial wave function and the interparticle spacing is also evident if we measure the particle separation probability density $\mathcal{P}(x_1 - x_2)$. This *one-dimensional* probability density is measured by recording the positions of each particle and finding the interparticle separation $(x_1 - x_2)$. To calculate this one-dimensional probability density, we integrate the two-particle probability density $\mathcal{P}(x_1, x_2)$ parallel to the $x_1 = x_2$ line (i.e., project the two-particle probability densities of Fig. 13.3 onto the diagonal line $x_2 = L - x_1$). The result of this calculation for the state with one particle in the $n = 1$ state and one particle in the $n = 2$ state of the infinite square well is shown in Fig. 13.4 (Problem 13.8). The distribution for symmetric spatial states is peaked at the origin, indicating that these identical particles are more likely than distinguishable particles to be found close to each other. The distribution for antisymmetric spatial states is zero at the origin (i.e., $x_1 = x_2$), indicating that the two identical particles cannot be found at the same location.

13.2.5 ■ Consequences of the Symmetrization Postulate

We have seen the effect of the symmetrization postulate on a two-particle system (an effective interaction that leads to changes in the interparticle spacing). The consequences of the symmetrization postulate for a many-particle system are much more radical, and are much different for systems of bosons and fermions. For example, in a three-particle system, the ground states are different because only two spin-1/2 fermions can be in the single-particle ground state, as illustrated in Figs. 13.5(a) and (b). For systems of N particles, the boson ground state has *all* N particles in the single-particle ground state and system energy NE_1 [Fig. 13.5(c)], while the spin-1/2 fermion ground state has energy levels occupied up to the $N/2$ single-particle state and system energy $\gg NE_1$ [Fig. 13.5(d)]. The proper study of these types of systems requires statistical mechanics and thermodynamics. The states depicted in Fig. 13.5 require temperatures near absolute zero so that the thermal energy is much less than the energy spacings.

For a system of bosons, if the requisite low temperature is reached and the density of the particles is high enough, then the ground state of the system exhibits a wealth of interesting quantum mechanical

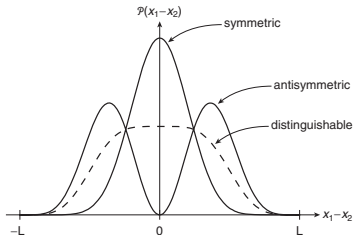


FIGURE 13.4 Probability density of interparticle separation for the first excited state ($n_a = 1$, $n_b = 2$) of a system of two particles in the infinite square well for the cases of distinguishable particles (dashed line), identical particles in symmetric spatial states (peaked at zero) and identical particles in antisymmetric spatial states (minimum at zero).

effects. When the inter-particle spacing is comparable to the de Broglie wavelength of the particles, then the system of bosons begins to behave as a single macroscopic quantum object. As the critical value of low temperature and high density is reached, the quantum mechanical attraction of the bosons arising from the symmetrization postulate takes over and the system “collapses” into the ground state. This dramatic event is a phase transition in the state of the matter and is called **Bose-Einstein condensation**. Liquid helium exhibits this phase transition at 2.18 K. The specific heat and the thermal conductivity increase discontinuously to signal the onset of the Bose-Einstein condensation. The viscosity of liquid helium drops dramatically and the system behaves as a **superfluid**, easily flowing through small capillaries and even up and out of its container. Liquid helium is a strongly interact-

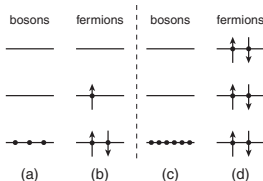


FIGURE 13.5 Ground states of multiple particle systems in a one-dimensional potential for a three-particle system of (a) bosons or (b) spin-1/2 fermions, and a six-particle system of (c) bosons or (d) spin-1/2 fermions.

ing system, so the theory of its low temperature quantum behavior is quite complicated. Dilute atomic gases provide a better testing ground for the study of the basic quantum mechanics of Bose-Einstein condensation. Recent experiments have cooled atoms to temperatures below 1 μK and achieved Bose-Einstein condensation. The atoms are close enough to have overlapping de Broglie wavelengths, but far enough apart that the atomic interactions are small. Moreover, the strength of the interactions can be adjusted through magnetic field changes, and the quantum effects can be studied as a function of the strength of the interaction. This new field has spawned a wealth of interesting effects and garnered the Nobel Prize in Physics in 2001.

For fermions, the behavior of a multiparticle system is dominated by the particles near the highest occupied state. The fermions at the low energy levels are “buried” in the sea of fermions and have nowhere to go, because the Pauli exclusion principle forbids them from making transitions to states that are already occupied. Only particles near the top of the distribution see nearby unoccupied levels to which they might make transitions. For example, in atoms, the electrons near the top are the valence electrons that determine the spectroscopy and chemistry of the atom. For electrons in solids, the energy at the top of the distribution of fermions is called the **Fermi energy** and plays a vital role in the behavior of the solid.

13.3 ■ INTERACTING PARTICLES

The apparent spatial “attraction” or “repulsion” of identical particles evident in Eq. (13.39) and Fig. 13.4 has a profound effect when we consider a real interaction between the two particles. The different spatial correlations of the particles lead to different energy shifts for the different spatial symmetry states. Consider two particles in a one-dimensional potential energy well, with an interaction potential energy between the two particles. We assume that this new term is small enough that we can treat it with perturbation theory. Assume that this interaction potential energy depends only on the particle separation:

$$H' = V_{int}(x_1 - x_2). \quad (13.40)$$

We use perturbation theory to find the first-order energy corrections

$$E^{(1)} = \langle \psi^{(0)} | H' | \psi^{(0)} \rangle \quad (13.41)$$

using the states from the last section as the zeroth-order states. We assume that the interaction is spin independent, so the spin does not affect this perturbation calculation. That is, in the matrix element

$$\begin{aligned} \langle \psi^{(0)} | H' | \psi^{(0)} \rangle &= \langle \psi_{spatial} | \langle \psi_{spin} | H' | \psi_{spatial} \rangle | \psi_{spin} \rangle \\ &= \langle \psi_{spatial} | H' | \psi_{spatial} \rangle \langle \psi_{spin} | \psi_{spin} \rangle \\ &= \langle \psi_{spatial} | H' | \psi_{spatial} \rangle, \end{aligned} \quad (13.42)$$

the spatial and spin states separate and only the spatial states enter into the perturbation calculation. The only role of the spin is to determine the allowed spatial states through the symmetrization postulate.

For a system of two identical spin-0 bosons, the zeroth-order ground state is

$$| \psi_{11}^{ss} \rangle \doteq \psi_{11}^s(x_1, x_2) | 00 \rangle = \varphi_1(x_1) \varphi_1(x_2) | 00 \rangle, \quad (13.43)$$

so the first-order perturbation is

$$\begin{aligned}
 E_{11}^{(1)} &= \langle \psi_{11}^S | H' | \psi_{11}^S \rangle \langle 00 | 00 \rangle \\
 &= \langle \psi_{11}^S | V_{int}(x_1 - x_2) | \psi_{11}^S \rangle \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi_1^*(x_1) \varphi_1^*(x_2) V_{int}(x_1 - x_2) \varphi_1(x_1) \varphi_1(x_2) dx_1 dx_2 \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\varphi_1(x_1)|^2 V_{int}(x_1 - x_2) |\varphi_1(x_2)|^2 dx_1 dx_2.
 \end{aligned} \tag{13.44}$$

It is convenient to define the general form of this matrix element as the **direct integral**

$$J_{nm} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\varphi_n(x_1)|^2 V_{int}(x_1 - x_2) |\varphi_m(x_2)|^2 dx_1 dx_2. \tag{13.45}$$

The direct integral is the interaction energy between the two probability densities $\mathcal{P}_n(x_1) = |\varphi_n(x_1)|^2$ and $\mathcal{P}_m(x_2) = |\varphi_m(x_2)|^2$ that represent the two particles. With this definition, the perturbed ground-state energy for a system of two spin-0 bosons is

$$E_{11} = 2E_1^{(0)} + J_{11}. \tag{13.46}$$

For a system of two identical spin-1/2 fermions, the zeroth-order ground state is

$$|\psi_{11}^{SA}\rangle \doteq \psi_{11}^S(x_1, x_2) |00\rangle = \varphi_1(x_1) \varphi_1(x_2) |00\rangle \tag{13.47}$$

and the first-order perturbation is

$$\begin{aligned}
 E_{11}^{(1)} &= \langle \psi_{11}^S | H' | \psi_{11}^S \rangle \langle 00 | 00 \rangle \\
 &= \langle \psi_{11}^S | V_{int}(x_1 - x_2) | \psi_{11}^S \rangle \\
 &= J_{11},
 \end{aligned} \tag{13.48}$$

which is the same as the boson case. The interaction is spin independent, and the ground-state spatial wave function is the same for bosons and fermions.

For the first excited state of the two-particle system, the identical spin-0 bosons must have a symmetric wave function [Eq. (13.26)], so the state vector is

$$|\psi_{12}^{SS}\rangle \doteq \psi_{12}^S(x_1, x_2) |00\rangle = \frac{1}{\sqrt{2}} [\varphi_1(x_1) \varphi_2(x_2) + \varphi_1(x_2) \varphi_2(x_1)] |00\rangle \tag{13.49}$$

and the first-order perturbation is

$$\begin{aligned}
 E_{12}^{(1)} &= \langle \psi_{12}^S | H' | \psi_{12}^S \rangle \langle 00 | 00 \rangle \\
 &= \langle \psi_{12}^S | V_{int}(x_1 - x_2) | \psi_{12}^S \rangle \\
 &= \frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [\varphi_1^*(x_1) \varphi_2^*(x_2) + \varphi_1^*(x_2) \varphi_2^*(x_1)] V_{int}(x_1 - x_2) \\
 &\quad [\varphi_1(x_1) \varphi_2(x_2) + \varphi_1(x_2) \varphi_2(x_1)] dx_1 dx_2.
 \end{aligned} \tag{13.50}$$

This gives four terms, but they are equal in pairs if we swap the integration dummy variables x_1 and x_2 , yielding (Problem 13.10)

$$E_{12}^{(1)} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\varphi_1(x_1)|^2 V_{int}(x_1 - x_2) |\varphi_2(x_2)|^2 dx_1 dx_2 \\ + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi_1^*(x_1) \varphi_2^*(x_2) V_{int}(x_1 - x_2) \varphi_1(x_2) \varphi_2(x_1) dx_1 dx_2. \quad (13.51)$$

The first term in Eq. (13.51) is the direct integral J_{12} defined in Eq. (13.45). The second term is a new term, which we call the **exchange integral** and define, in general, as

$$K_{nm} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi_n^*(x_1) \varphi_m^*(x_2) V_{int}(x_1 - x_2) \varphi_n(x_2) \varphi_m(x_1) dx_1 dx_2. \quad (13.52)$$

With this definition, the energy of the first excited state of the system of two identical spin-0 bosons is

$$E_{12} = E_1^{(0)} + E_2^{(0)} + J_{12} + K_{12}. \quad (13.53)$$

The exchange integral has no classical explanation. It is a manifestation of the symmetrization requirement. It is not caused by spin, but it is intimately related to spin because of the role of spin in the symmetrization postulate.

For two identical spin-1/2 fermions, the excited state spatial wave function can be either symmetric or antisymmetric depending on the spin state. For the antisymmetric singlet spin state, the spatial wave function must be symmetric

$$|\psi_{12}^{SA}\rangle \doteq \psi_{12}^S(x_1, x_2)|00\rangle = \frac{1}{\sqrt{2}}[\varphi_1(x_1)\varphi_2(x_2) + \varphi_1(x_2)\varphi_2(x_1)]|00\rangle. \quad (13.54)$$

The first-order energy shift is

$$E_{12}^{(1)} = \langle \psi_{12}^S | H' | \psi_{12}^S \rangle \langle 00 | 00 \rangle \\ = \langle \psi_{12}^S | V_{int}(x_1 - x_2) | \psi_{12}^S \rangle \\ = J_{12} + K_{12}. \quad (13.55)$$

This is the same shift as the boson excited state because the spatial wave function is the same and the spin does not affect the expectation value.

For the symmetric triplet spin state, the spatial wave function is antisymmetric

$$|\psi_{12}^{AS}\rangle \doteq \psi_{12}^A(x_1, x_2)|1M\rangle = \frac{1}{\sqrt{2}}[\varphi_1(x_1)\varphi_2(x_2) - \varphi_1(x_2)\varphi_2(x_1)]|1M\rangle. \quad (13.56)$$

The resultant first-order energy correction

$$E_{12}^{(1)} = \langle \psi_{12}^A | H' | \psi_{12}^A \rangle \langle 1M | 1M \rangle \\ = \langle \psi_{12}^A | V_{int}(x_1 - x_2) | \psi_{12}^A \rangle \\ = J_{12} - K_{12} \quad (13.57)$$

has a negative exchange integral contribution because of the minus sign in the asymmetric spatial wave function (Problem 13.12). We combine the results in Eqs. (13.55) and (13.57) to express the energy of the first excited state of the two spin-1/2 fermion system as

$$E_{12} = E_1^{(0)} + E_2^{(0)} + J_{12} \pm K_{12}, \quad (13.58)$$

where the $+$ ($-$) sign refers to the symmetric (antisymmetric) spatial state and the respective singlet (triplet) state.

The energies of the ground and excited states are shown in Fig. 13.6, where we assume that the direct and exchange integrals J and K are positive, which is typical for the Coulomb interaction between identical charged particles. The direct integral raises all energy states because of the positive repulsive interaction expected classically for charged particles of the same sign. The exchange integral reflects the additional repulsive interaction caused by the spatial correlation or anticorrelation of the particles arising from the symmetrization postulate. For bosons and spin-singlet fermions, the spatial “attraction” that arises from the symmetrization postulate [Fig. 13.3(c)] increases the positive repulsive interaction energy because they are closer together in space. For spin-triplet fermions, the spatial “repulsion” that arises from the symmetrization postulate [Fig. 13.3(d)] decreases the interaction energy. The degeneracy of the excited state in the fermion case is partially lifted by the exchange

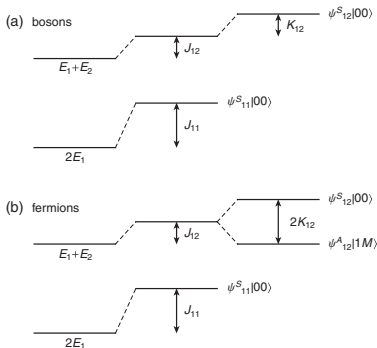


FIGURE 13.6 Energies and state vectors for the ground and first excited states of two identical (a) spin-0 bosons or (b) spin-1/2 fermions in a one-dimensional potential energy well.

integral term. The resultant energies depend on the spin of the system even though spin is not part of the interaction Hamiltonian. The spin plays its role by determining which spatial states are allowed.

13.4 ■ EXAMPLE: THE HELIUM ATOM

The symmetrization postulate and the resultant Pauli exclusion principle are key elements in understanding atomic structure and the periodic table. The ramifications of the symmetrization postulate are first evident in the case of the helium atom with two electrons. The helium Hamiltonian is similar to hydrogen but with added potential energy terms due to the second electron interacting with the doubly charged nucleus and the two electrons interacting with each other. The helium Hamiltonian is

$$H = \left(\frac{p_1^2}{2m} - \frac{2e^2}{4\pi\epsilon_0 r_{11}} \right) + \left(\frac{p_2^2}{2m} - \frac{2e^2}{4\pi\epsilon_0 r_{22}} \right) + \frac{e^2}{4\pi\epsilon_0 r_{12}}, \quad (13.59)$$

where r_{12} is the separation of the two electrons, as shown in Fig. 13.7. The Coulomb repulsion term between the two electrons is clearly of the same order of magnitude as the Coulomb terms representing the interaction of each electron with the nucleus, but we treat it as a perturbation so that we can write the Hamiltonian as a zeroth-order term whose solutions we know, plus a perturbation:

$$\begin{aligned} H &= H_0 + H' \\ H' &= \frac{e^2}{4\pi\epsilon_0 r_{12}}. \end{aligned} \quad (13.60)$$

The zeroth-order Hamiltonian is the sum of two hydrogen atom Hamiltonians, each with a nuclear charge $Z = 2$. The eigenstates and eigenenergies of a hydrogenic atom with nuclear charge Z are obtained from the hydrogen atom solutions from Chapter 8 by the substitution $e^2 \rightarrow Ze^2$, which scales the energies by a factor Z^2 and the size of the radial wave function by $1/Z$. For example, the ground-state wave function of a hydrogenic atom with nucleus $+Ze$ is

$$\psi_{100}(r, \theta, \phi) = \sqrt{\frac{Z^3}{\pi a_0^3}} e^{-Zr/a_0}, \quad (13.61)$$

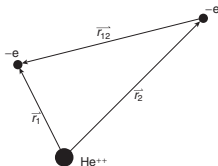


FIGURE 13.7 Helium atom coordinates.

where $a_0 = 4\pi\epsilon_0\hbar^2/me^2$ is the Bohr radius. The zeroth-order energy of the helium atom with one electron in state n_a and one electron in state n_b is the sum of the hydrogenic energies:

$$\begin{aligned} E_{n_a n_b}^{(0)} &= -Z^2 \text{Ryd} \left(\frac{1}{n_a^2} + \frac{1}{n_b^2} \right) \\ &= -4 \text{Ryd} \left(\frac{1}{n_a^2} + \frac{1}{n_b^2} \right). \end{aligned} \quad (13.62)$$

Though we don't expect this perturbation approach to yield very precise results, it is a useful first attempt and illustrates many of the important new aspects that arise from the symmetrization postulate.

As we did in the previous section, we separate the spin and spatial aspects of the state vectors because spin and position are not coupled. The complete eigenstates have the form

$$|\psi\rangle = |\psi_{\text{spatial}}\rangle |\psi_{\text{spin}}\rangle \quad (13.63)$$

and must be antisymmetric under exchange of the two fermions. Thus the spatial part must be symmetric if the spin part is antisymmetric, or the spatial part must be antisymmetric if the spin part is symmetric.

The spatial part of the state vector represents the state with one particle in the hydrogenic state $n_a\ell_a m_a$ and one particle in the state $n_b\ell_b m_b$. The properly symmetrized spatial wave functions are

$$\begin{aligned} \psi_{n_a\ell_a m_a, n_b\ell_b m_b}^S(\mathbf{r}_1, \mathbf{r}_2) &= \frac{1}{\sqrt{2}} [\psi_{n_a\ell_a m_a}(\mathbf{r}_1) \psi_{n_b\ell_b m_b}(\mathbf{r}_2) + \psi_{n_a\ell_a m_a}(\mathbf{r}_2) \psi_{n_b\ell_b m_b}(\mathbf{r}_1)] \\ \psi_{n_a\ell_a m_a, n_b\ell_b m_b}^A(\mathbf{r}_1, \mathbf{r}_2) &= \frac{1}{\sqrt{2}} [\psi_{n_a\ell_a m_a}(\mathbf{r}_1) \psi_{n_b\ell_b m_b}(\mathbf{r}_2) - \psi_{n_a\ell_a m_a}(\mathbf{r}_2) \psi_{n_b\ell_b m_b}(\mathbf{r}_1)]. \end{aligned} \quad (13.64)$$

The spin part of the state vector is obtained by properly symmetrizing the spin states of two spin-1/2 fermions, which yields the eigenstates $|SM\rangle$ of the total spin with $S = 0$ or 1 . The four states are

$$\left. \begin{aligned} |11\rangle &= |++\rangle \\ |10\rangle &= \frac{1}{\sqrt{2}}[|+-\rangle + |-+\rangle] \\ |1,-1\rangle &= |--\rangle \\ |00\rangle &= \frac{1}{\sqrt{2}}[|+-\rangle - |-+\rangle] \end{aligned} \right\} \begin{array}{l} \text{Triplet state} \\ \text{Singlet state} \end{array} \quad (13.65)$$

The complete antisymmetric quantum state vector of the helium atom is obtained by combining the antisymmetric singlet state with symmetric spatial wave function or by combining the symmetric triplet state with antisymmetric spatial wave function. Thus, the only possible states are:

$$\begin{aligned} |\psi_{n_a\ell_a m_a, n_b\ell_b m_b}^{SA}\rangle &= |\psi_{n_a\ell_a m_a, n_b\ell_b m_b}^S\rangle |00\rangle \\ |\psi_{n_a\ell_a m_a, n_b\ell_b m_b}^{AS}\rangle &= |\psi_{n_a\ell_a m_a, n_b\ell_b m_b}^A\rangle |1M\rangle. \end{aligned} \quad (13.66)$$

The other combinations are not possible states for this system.

13.4.1 ■ Helium Ground State

The ground state of helium has both electrons in hydrogenic ground states, so the antisymmetric spatial state is identically zero [Eq. (13.22)] and only the symmetric spatial state is allowed. To ensure

that the total state vector is antisymmetric, Eq. (13.66) tells us that the spin part of the ground state must be the antisymmetric singlet state $|00\rangle$. The triplet state $|1M\rangle$ is not permitted in the helium ground state. Thus the ground state of helium is

$$|\psi_{\text{ground}}\rangle = |\psi_{1s,1s}^{\text{SA}}\rangle = |\psi_{1s,1s}^{\text{S}}\rangle|00\rangle, \quad (13.67)$$

with a zeroth-order energy determined by the sum of two hydrogenic ground-state energies:

$$E_{1s,1s}^{(0)} = -4 \text{ Ryd} \left(\frac{1}{1^2} + \frac{1}{1^2} \right) = -8 \text{ Ryd} = -108.8 \text{ eV}. \quad (13.68)$$

The zeroth-order helium energy states are shown in Fig. 13.8, obtained using Eq. (13.62).

An aside about energy levels is in order here. In hydrogen, the ground-state energy is -13.6 eV , where zero energy corresponds to the electron and proton infinitely far apart and at rest. We refer to this zero energy level as the **ionization level**. For the calculation we have just done for the helium ground state, the zero of energy corresponds to both electrons removed to infinity, and so is referred to as the **double ionization level**. However, it is more common in the literature to quote atomic energy

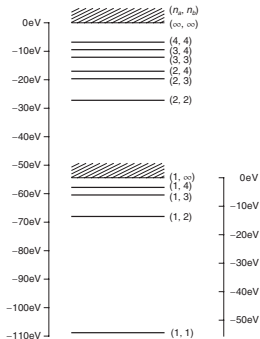


FIGURE 13.8 Helium atom energies in zeroth order, where hydrogenic Bohr energies are assumed. The energy scale on the left is referenced to the double ionization level and the energy scale on the right is referenced to the single ionization level, which is at -54.4 eV with respect to the double ionization level.

levels with respect to the **single ionization level** corresponding to one electron removed from the atom. If we remove one electron from helium, we are left with a hydrogenic ion with an energy

$$E_{1s,\infty}^{(0)} = -4 \text{ Ryd} \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) = -4 \text{ Ryd} = -54.4 \text{ eV}, \quad (13.69)$$

which is half of the energy in Eq. (13.68). To quote energies referenced to the single ionization level, we must subtract this energy (as shown in Fig. 13.8), in which case we get a helium ground-state energy of -54.4 eV .

The experimental value for the helium ground-state energy is -25 eV (referenced to the single ionization level), which is quite different from our zeroth-order estimate. This is not unexpected, as we said above that the electron-electron interaction, which is neglected in zeroth-order, is the same order of magnitude as the electron-nuclear interactions that are responsible for the binding. Even though they are the same order, our approach is to treat the electron-electron repulsion as a perturbation and find the perturbed energies of this system. The helium ground state is nondegenerate, so we find the shift caused by the perturbation by finding the expectation value of the perturbation in the zeroth-order state:

$$\begin{aligned} E_{1s,1s}^{(1)} &= \langle \psi_{1s,1s}^{\text{SA}} | H' | \psi_{1s,1s}^{\text{SA}} \rangle \\ &= \langle \psi_{1s,1s}^S | \langle 00 | \frac{e^2}{4\pi\epsilon_0 r_{12}} | \psi_{1s,1s}^S \rangle | 00 \rangle \\ &= \langle \psi_{1s,1s}^S | \frac{e^2}{4\pi\epsilon_0 r_{12}} | \psi_{1s,1s}^S \rangle \langle 00 | 00 \rangle \\ &= \iint \psi_{100}^*(\mathbf{r}_1) \psi_{100}^*(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} \psi_{100}(\mathbf{r}_2) \psi_{100}(\mathbf{r}_1) d^3\mathbf{r}_1 d^3\mathbf{r}_2. \end{aligned} \quad (13.70)$$

This integral is the direct integral we defined in the one-dimensional example in Eq. (13.45). In this three-dimensional Coulomb interaction problem, we define the direct integral as

$$J_{n\ell,n'\ell'} = \iint |\psi_{n\ell m}(\mathbf{r}_1)|^2 \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} |\psi_{n'\ell'm'}(\mathbf{r}_2)|^2 d^3\mathbf{r}_1 d^3\mathbf{r}_2. \quad (13.71)$$

These integrals are independent of m , but not ℓ , which is why we drop the m subscript on the energies. To calculate the direct integrals, it is useful to use the spherical harmonic addition theorem

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} Y_{\ell m}(\theta_1, \phi_1) Y_{\ell m}(\theta_2, \phi_2), \quad (13.72)$$

where $r_{>}$ stands for the larger of the two distances r_1 and r_2 , and $r_{<}$ the smaller.

The ground-state direct integral in Eq. (13.70) can be done and the result is (Problem 13.14):

$$E_{1s,1s}^{(1)} = \frac{5}{8} \frac{Ze^2}{4\pi\epsilon_0 a_0} = \frac{5}{4} \frac{e^2}{4\pi\epsilon_0 a_0} = \frac{10}{4} \text{ Ryd} = 34 \text{ eV}. \quad (13.73)$$

The shift is positive because the electrons repel each other, yielding a positive Coulomb potential energy. The new estimate of the ground-state energy (relative to the single ionization level) is

$$E_{1s,1s} \equiv E_{1s,1s}^{(0)} + E_{1s,1s}^{(1)} = -54.4 \text{ eV} + 34 \text{ eV} = -20.4 \text{ eV}, \quad (13.74)$$

which is now much closer to the experimental value of -25 eV . To make a better estimate, we would have to account for the shielding of the nuclear charge by the presence of the second electron.

13.4.2 ■ Helium Excited States

Now let's turn our attention to the excited states of helium. The zeroth-order energy level diagram in Fig. 13.8 makes it clear that all states with both electrons excited have a zeroth-order energy above the single ionization level $E_{1s,\infty}^{(0)}$. For example, the doubly excited state $n_a = 2, n_b = 2$ has an energy

$$E_{2,2}^{(0)} = -4 \text{ Ryd} \left(\frac{1}{2^2} + \frac{1}{2^2} \right) = -2 \text{ Ryd} = -27.2 \text{ eV}, \quad (13.75)$$

which is 27.2 eV above the single ionization level. Such doubly excited states are not stable. They decay to a lower energy state with one electron in the hydrogenic ground state and the second electron traveling to infinity with the excess energy. This decay is very likely and so the lifetime of the doubly excited states is very short. The likelihood of this process leads to its name: **auto-ionization**. For this reason, it is common to limit the discussion of excited atomic states (in this helium example as well as other atomic systems) to those where only one electron is excited and the others remain in the atomic ground state.

Because the excited electron is in a different spatial state than the remaining ground-state electron, both the symmetric and antisymmetric spatial states are allowed. We also expect additional degeneracy because the hydrogen excited states are degenerate with respect to the angular momentum quantum numbers ℓ and m .

The first excited state of helium has $n_a = 1, n_b = 2$. The two possible states are:

$$\begin{aligned} |\psi_{1s,2\ell}^{\text{SA}}\rangle &= |\psi_{1s,2\ell}^{\text{S}}\rangle|00\rangle \doteq \frac{1}{\sqrt{2}}[\psi_{100}(\mathbf{r}_1)\psi_{2\ell m}(\mathbf{r}_2) + \psi_{100}(\mathbf{r}_2)\psi_{2\ell m}(\mathbf{r}_1)]|00\rangle \\ |\psi_{1s,2\ell}^{\text{AS}}\rangle &= |\psi_{1s,2\ell}^{\text{A}}\rangle|1M\rangle \doteq \frac{1}{\sqrt{2}}[\psi_{100}(\mathbf{r}_1)\psi_{2\ell m}(\mathbf{r}_2) - \psi_{100}(\mathbf{r}_2)\psi_{2\ell m}(\mathbf{r}_1)]|1M\rangle. \end{aligned} \quad (13.76)$$

In both the symmetric and antisymmetric spatial cases, there are four possible states corresponding to the single $2s$ ($\ell = 0, m = 0$) and the three $2p$ ($\ell = 1, m = 0, \pm 1$) states. When we combine these states with the single spin singlet state and the three spin triplet states, we find that there are 16 possible states overall. All these states are degenerate in the unperturbed system with Hamiltonian H_0 .

The unperturbed energy of these states is the hydrogenic energy shown in Fig. 13.8. We apply degenerate perturbation theory to find the effect of the electron-electron repulsion term H' on these 16 degenerate states. The perturbation Hamiltonian is diagonal, so the energy corrections are the diagonal elements

$$E_{1s,2\ell}^{(1)} = \langle \psi_{1s,2\ell}^{\text{SA}} | H' | \psi_{1s,2\ell}^{\text{SA}} \rangle \quad (13.77)$$

for the symmetric spatial state, and

$$E_{1s,2\ell}^{(1)} = \langle \psi_{1s,2\ell}^{\text{AS}} | H' | \psi_{1s,2\ell}^{\text{AS}} \rangle \quad (13.78)$$

for the antisymmetric spatial state. In both cases, the spin states are unaffected by the perturbation [see Eq. (13.70)] so we are left with a spatial integral:

$$\begin{aligned} E_{1s,2\ell}^{(1)} &= \iint \frac{1}{\sqrt{2}} [\psi_{100}^*(\mathbf{r}_1)\psi_{2\ell m}^*(\mathbf{r}_2) \pm \psi_{100}(\mathbf{r}_2)\psi_{2\ell m}(\mathbf{r}_1)] \frac{e^2}{4\pi\epsilon_0|\mathbf{r}_1 - \mathbf{r}_2|} \\ &\quad \frac{1}{\sqrt{2}} [\psi_{100}^*(\mathbf{r}_1)\psi_{2\ell m}^*(\mathbf{r}_2) \pm \psi_{100}(\mathbf{r}_2)\psi_{2\ell m}(\mathbf{r}_1)] d^3\mathbf{r}_1 d^3\mathbf{r}_2, \end{aligned} \quad (13.79)$$

where the $\pm$ distinguishes the two states in Eq. (13.76). This gives four terms, but they are equal in pairs if we swap the integration dummies $\mathbf{r}_1$ and $\mathbf{r}_2$. Hence, we cancel the factor of 1/2 and get two terms:

$$E_{1s,2\ell}^{(1)} = \iint |\psi_{100}(\mathbf{r}_1)|^2 \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} |\psi_{2\ell m}(\mathbf{r}_2)|^2 d^3\mathbf{r}_1 d^3\mathbf{r}_2 \\ \pm \iint \psi_{100}^*(\mathbf{r}_1) \psi_{2\ell m}^*(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} \psi_{100}(\mathbf{r}_2) \psi_{2\ell m}(\mathbf{r}_1) d^3\mathbf{r}_1 d^3\mathbf{r}_2. \quad (13.80)$$

The first term is the direct integral and the second term is the exchange integral, which in general is

$$K_{n\ell, n'\ell'} = \iint \psi_{n\ell m}^*(\mathbf{r}_1) \psi_{n'\ell' m'}^*(\mathbf{r}_2) \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} \psi_{n\ell m}(\mathbf{r}_2) \psi_{n'\ell' m'}(\mathbf{r}_1) d^3\mathbf{r}_1 d^3\mathbf{r}_2. \quad (13.81)$$

So we write the energy perturbation as

$$E_{1s,2\ell}^{(1)} = J_{1s,2\ell} \pm K_{1s,2\ell}, \quad (13.82)$$

where the $+$ ($-$) sign refers to the symmetric (antisymmetric) spatial state and the respective singlet (triplet) state. The direct integral is also called the Coulomb interaction energy because it is the electrostatic interaction potential energy of the two electrons: one in the $1s$ state and the other in the $2s$ or $2p$ state.

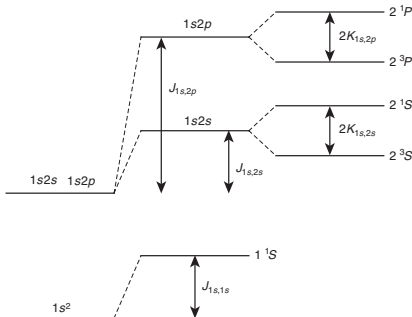


FIGURE 13.9 Shifts and splittings of the helium ground state and first excited state caused by the direct and exchange interactions.

Both the direct and exchange integrals in helium are positive, so the singlet states are higher in energy than the corresponding triplet states, as shown in Fig. 13.9. The energy levels are labeled with a modified spectroscopic notation $n^{2S+1}L$ [see Eq. (11.84)], with n being the state of the excited electron. The J label is suppressed because it has no bearing on the energy at this order of approximation. We understand the singlet-triplet ordering of the energy levels by noting that in the singlet state, the spin state is antisymmetric and the spatial state is symmetric, implying that the two electrons get closer to each other and therefore increase the repulsive Coulomb potential energy. In the triplet state, the spin state is symmetric, the spatial state is antisymmetric, and the two electrons are farther apart, thus lowering the Coulomb potential energy.

This ordering of the energy levels, with the singlet state above the triplet state, is evident throughout the excited states of helium. Another important feature of the singlet and triplet states is that optical transitions between these states are forbidden. The electromagnetic light field does not couple to the spin, so the selection rules for optical transitions require there to be no change in the spin quantum number between two states. Hence, transitions between the singlet and triplet states of helium are forbidden, and it was originally believed that there were two types of helium: parahelium ($S = 0$) and orthohelium ($S = 1$). Thus, energy diagrams of helium often show the singlet and triplet levels separately, as in Fig. 13.10. We now know that transitions between parahelium and orthohelium do occur, with small probability, due to higher-order effects. Note that the 2^3S state of orthohelium is the lowest state on the triplet side. Due to the spin selection rule, it is metastable against decay to the ground state 1^1S . The lifetime of this metastable state is 8000 seconds, which is generally much longer than the time it takes a helium atom to travel through an experimental system, so the state effectively has an infinite lifetime in laboratory experiments.

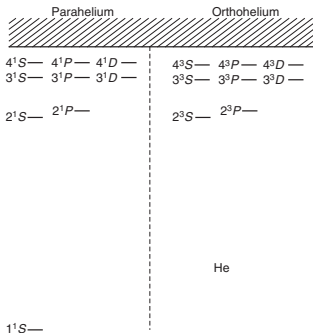


FIGURE 13.10 Helium energy spectrum.

13.5 ■ THE PERIODIC TABLE

The helium atom illustrates the importance of the symmetrization postulate in determining the spectrum of energy levels of a multielectron atom. Let's now qualitatively explain how the symmetrization postulate, in the guise of the Pauli exclusion principle, determines the structure of the periodic table as the atomic number Z increases. To a zeroth approximation, the states of multielectron atoms are the hydrogenic states labeled with n , ℓ , and m . However, we must also include spin, so there are four quantum numbers n , ℓ , m_ℓ , and m_s labeling each electron (the fifth number $s = 1/2$ is the same for all electrons so we suppress it). The zeroth-order hydrogen energy states E_n depend only on the quantum number n and are n^2 degenerate with respect to ℓ and m_ℓ . The additional m_s degree of freedom doubles the degeneracy, so that each hydrogenic energy level is $2n^2$ degenerate. We refer to each n energy level as a **shell** and to each $n\ell$ orbital as a **subshell**. Each subshell has $2(2\ell + 1)$ possible states.

If electron were bosons, any number could occupy the hydrogenic ground state $n = 1$, similar to Fig. 13.5(c), and chemistry would be boring. But because electrons are fermions, only one electron can occupy each state specified by the four quantum numbers n , ℓ , m_ℓ , and m_s . Hence, as the atomic number Z increases through the periodic table, we expect that each additional electron occupies the lowest available hydrogenic energy state, filling each subshell with $2(2\ell + 1)$ electrons and each shell with $2n^2$ electrons, analogous to Fig. 13.5(d). The resulting electronic configurations are denoted by listing the subshells with the number of electrons in each subshell as a superscript. (e.g., $1s^2$). We thus expect the periodic table to reflect the pyramidal structure shown in Table 13.1. But that would mean that the seven rows of the periodic table would have 280 atoms, whereas we know there are just over 100 atoms. The periodic table does have a pyramidal structure, but not one that reflects the numbers in Table 13.1. Why not?

The primary reason is that the nuclear charge is shielded by inner shell electrons in a way that lifts the ℓ degeneracy we expect from the simple hydrogen case, giving rise to energy levels specified by the n and ℓ quantum numbers. For a given n , higher values of ℓ correspond to orbits farther from the nucleus because the increased angular momentum leads to a larger centrifugal barrier. Hence, the electrons in high angular momentum orbitals are shielded from the nuclear attraction by the electrons in lower orbits and are bound less tightly. This screening effect explains why electrons fill the hydrogenic orbitals in the sequence $1s$, $2s$, $2p$, $3s$, etc. However, the screening effect is so large that it exceeds the hydrogenic $n \rightarrow n + 1$ level separation in some cases, which disturbs the expected shell filling structure of Table 13.1. A schematic of the ordering of the energy levels of multielectron atoms is shown

Table 13.1 Electronic Configurations in a Periodic Table Based Upon Purely Hydrogenic Energy Levels

Shell (n)	Subshell Configuration	Degeneracy ($2n^2$)
1	$1s^2$	2
2	$2s^2 2p^6$	8
3	$3s^2 3p^6 3d^{10}$	18
4	$4s^2 4p^6 4d^{10} 4f^{14}$	32
5	$5s^2 5p^6 5d^{10} 5f^{14} 5g^{18}$	50
6	$6s^2 6p^6 6d^{10} 6f^{14} 6g^{18} 6h^{22}$	72
7	$7s^2 7p^6 7d^{10} 7f^{14} 7g^{18} 7h^{22} 7i^{26}$	98

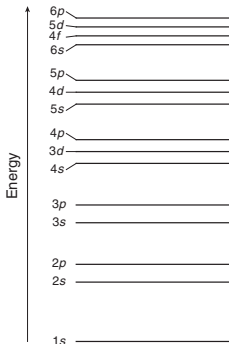


FIGURE 13.11 Approximate ordering of the energies of subshells after accounting for the shielding of the nuclear charge. The energies are not to scale.

in Fig. 13.11. The screening effect results in four major differences from the unshielded hydrogenic model: (1) the ℓ degeneracy is lifted, (2) the nd levels are shifted up to lie above the $(n + 1)s$ levels, (3) the nf levels are shifted up to lie above the $(n + 2)s$ levels, and (4) the np levels are the highest levels within their “group” of levels. Hence the energy filling proceeds in the manner shown in Table 13.2, with the number of atoms per row shown at right.

Table 13.2 Electronic Configurations in the Periodic Table

Row	Subshell Configuration				Number of Atoms
1	$1s^2$				2
2	$2s^2$ $2p^6$				8
3	$3s^2$ $3p^6$				8
4	$4s^2$	$3d^{10}$	$4p^6$		18
5	$5s^2$	$4d^{10}$	$5p^6$		18
6	$6s^2$	$4f^{14}$	$5d^{10}$	$6p^6$	32
7	$7s^2$	$5f^{14}$	$6d^{10}$	$7p^6$	32

screening effect pushes the $3d$ energy level up near the $4s$ energy level, so the third row has only the $3s$ and $3p$ subshells. The $3d$ states are not filled until the fourth row of the periodic table. The $4s$ states are filled first for potassium and calcium, then the $3d$ states are filled for scandium through zinc, and finally the $4p$ states are filled for gallium through krypton. The $4s$ and $3d$ levels are so close that there are some anomalies in the transition metals in the fourth row, as indicated in Table 13.3. Chromium and copper each have only one $4s$ electron and one more $3d$ electron than you might expect. The fifth row fills in the order $5s$, $4d$, and $5p$, analogous to the fourth row because the f subshells are pushed up two groups. The fifth row transition metals also exhibit anomalies in the $5s$ and $4d$ ordering, with palladium being the most extreme in having no $5s$ electrons. The sixth and seventh rows both include f subshells and also have anomalous filling among the s , f , and d subshells.

13.6 ■ EXAMPLE: THE HYDROGEN MOLECULE

Now let's take a look at another two-electron system that will introduce us to some molecular physics and prepare us for the periodic systems in Chapter 15. Consider the hydrogen molecule with two nuclei (protons), each with a bound electron, with the two atoms bound to each other to make a four-particle system, as shown in Fig. 13.13. We label the electrons 1 and 2 and the protons A and B . The Hamiltonian for the molecule includes hydrogen Hamiltonians for each electron-proton pair; additional Coulomb potential energy terms for the electron-electron, proton-proton, and electron-other-proton pairs; and kinetic energy for the nuclei:

$$H = H_{atom,1A} + H_{atom,2B} + V_{ee} + V_{pp} + V_{ep} + T_N, \quad (13.83)$$

where

$$\begin{aligned} H_{atom,1A} &= \left(\frac{p_1^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r_{1A}} \right) \\ H_{atom,2B} &= \left(\frac{p_2^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r_{2B}} \right) \\ V_{ee} &= \frac{e^2}{4\pi\epsilon_0 r_{12}} \\ V_{pp} &= \frac{e^2}{4\pi\epsilon_0 R_{AB}} \\ V_{ep} &= -\frac{e^2}{4\pi\epsilon_0 r_{1B}} - \frac{e^2}{4\pi\epsilon_0 r_{2A}} \\ T_{nuc} &= \frac{P_A^2}{2M_A} + \frac{P_B^2}{2M_B}. \end{aligned} \quad (13.84)$$

We treat this two-electron system as we did the helium atom in the sense that we put the electrons into the lowest energy states of the one-electron system and then account for the required symmetrization of the two identical electrons. So we must first discuss the one-electron molecule.

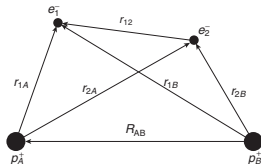


FIGURE 13.13 Hydrogen molecule.

13.6.1 ■ The Hydrogen Molecular Ion H_2^+

The hydrogen molecular ion H_2^+ has one electron and has a Hamiltonian

$$H_{\text{ion}} = \frac{p_1^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r_{1A}} - \frac{e^2}{4\pi\epsilon_0 r_{1B}} + \frac{e^2}{4\pi\epsilon_0 R_{AB}} + \frac{p_A^2}{2M_A} + \frac{p_B^2}{2M_B}. \quad (13.85)$$

We do not need the subscript labeling the electron as #1, but we keep it to connect with the H_2 case. To construct approximate energy eigenstates, we use the method of **linear combination of atomic orbitals** (LCAO), which assumes that we can use the atomic energy eigenstates as basis functions. If the two protons are far apart, then we expect that in the ground state of the ion, the electron is attached to one proton and is in the hydrogen atomic ground state. The electronic wave function in this case is

$$|\psi_{\text{separated}}\rangle \doteq \psi_{1s}(\mathbf{r}_{1A}), \quad (13.86)$$

assuming the electron is on proton A. However, the ion Hamiltonian in Eq. (13.85) is spatially symmetric about the center of the molecule located at the midpoint of the internuclear separation R_{AB} and the eigenstates should reflect this spatial symmetry. Hence we construct two possible ground states of the ion that are symmetric and antisymmetric spatially:

$$\begin{aligned} |\psi_{1s}^g\rangle_1 &\doteq \frac{1}{\sqrt{2}}[\psi_{1s}(\mathbf{r}_{1A}) + \psi_{1s}(\mathbf{r}_{1B})] \\ |\psi_{1s}^u\rangle_1 &\doteq \frac{1}{\sqrt{2}}[\psi_{1s}(\mathbf{r}_{1A}) - \psi_{1s}(\mathbf{r}_{1B})]. \end{aligned} \quad (13.87)$$

These states are even (*g*) and odd (*u*), respectively, under reflection about the midpoint of R_{AB} , and are labeled as *gerade* and *ungerade* states (German for even and odd). We use this labeling notation to distinguish the *spatial* symmetry (*g, u*) from the *exchange* symmetry (*S, A*) that we'll need for the H_2 two-electron molecule.

To estimate the ground-state energy of the ion, we calculate the expectation value of the energy $\langle E \rangle = \langle H_{\text{ion}} \rangle$ using the wave functions in Eq. (13.87). We ignore the motion of the nuclei by assuming that the internuclear separation R_{AB} is fixed. In calculating the energy expectation value, we integrate over the electron position, but the result is still dependent on the choice for the fixed value of R_{AB} . This dependence is evident in the results shown in Fig. 13.14. For both the *gerade* and *ungerade* states, the energy at large internuclear separation is simply the hydrogen energy -13.6 eV expected

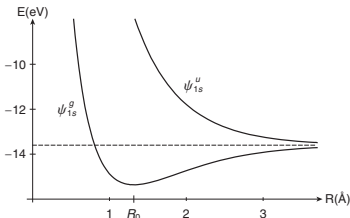


FIGURE 13.14 Energies of the bonding and antibonding orbitals of the hydrogen molecular ion, as a function of the internuclear separation.

for the system of one ground state atom and one distant proton. At very small internuclear separation ($R_{AB} \ll a_0$), the energy of both states becomes positive and very large due to the strong proton-proton Coulomb repulsion. For intermediate internuclear separation, the *gerade* and *ungerade* states have different energies. The minimum in the *gerade* state energy indicates an attraction that leads to a stable molecule with an internuclear separation given by the bond length R_0 . The energy of the *ungerade* state has no minimum and is repulsive at all distances, implying that a system in this state will dissociate into a bound hydrogen atom and an isolated proton. Hence, we refer to the *gerade* state as a **bonding orbital** and the *ungerade* state as an **antibonding orbital**. Note that the energy of the bonding orbital shown in Fig. 13.14 is the potential energy function we used in Chapter 9 (Fig. 9.14) to find the motion of the nuclei in a diatomic molecule. This approximate method of treating the electron motion first and then the nuclear motion is the **Born-Oppenheimer approximation**. It relies on the assumption that the nuclear motion is much slower than the electron motion because of the large mass difference.

To gain a qualitative understanding of the differences between the *gerade* and *ungerade* states, consider a one-dimensional view of the wave functions and probability densities of the two states. Along the line of the internuclear separation, the *gerade* and *ungerade* states are

$$\begin{aligned}
 |\psi_{1s}^g\rangle &\doteq \frac{1}{\sqrt{2}} \left[\psi_{1s} \left(r + \frac{R_{AB}}{2} \right) + \psi_{1s} \left(r - \frac{R_{AB}}{2} \right) \right] \\
 |\psi_{1s}^u\rangle &\doteq \frac{1}{\sqrt{2}} \left[\psi_{1s} \left(r + \frac{R_{AB}}{2} \right) - \psi_{1s} \left(r - \frac{R_{AB}}{2} \right) \right].
 \end{aligned}
 \tag{13.88}$$

Substituting the hydrogen atomic ground-state wave function $\psi_{1s}(r) = e^{-r/a_0}/\sqrt{\pi a_0^3}$ into Eq. (13.88) yields the plots shown in Fig. 13.15. For the *gerade* state, the wave functions add [Fig. 13.15(a)] and the resulting electron probability density [Fig. 13.15(b)] is large between the two protons. This excess negative charge increases the attractive Coulomb interaction of the electron and protons enough to overcome the proton-proton Coulomb repulsion and permit a stable bound molecule. In contrast, the wave functions of the *ungerade* state subtract [Fig. 13.15(c)] and produce a

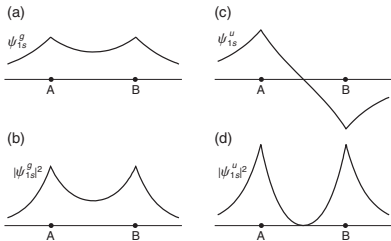


FIGURE 13.15 Hydrogen molecular ion wave functions (a, c) and probability densities (b, d) for *gerade* (a, b) and *ungerade* (c, d) states.

zero point in the electron density [Fig. 13.15(d)] between the two protons. This deficiency of negative charge between the protons causes the proton-proton Coulomb repulsion to dominate and leads to the antibonding behavior of the *ungerade* state.

13.6.2 ■ The Hydrogen Molecule H_2

We now return to the hydrogen molecule with two electrons. In the atomic case, our first guess for the ground state of the two-electron helium atom was to put both electrons in the $1s$ ground hydrogenic atomic state (with $Z = 2$) in a symmetric spatial state and to form an antisymmetric spin-singlet state to satisfy the symmetrization postulate. By analogy, our first guess for the ground state of the two-electron hydrogen molecule puts each electron in the $|\psi_{1s}^g\rangle$ ground hydrogen molecular ion state to make a spatial state that is symmetric with respect to exchange and puts the two electrons in an antisymmetric spin-singlet state to satisfy the symmetrization postulate:

$$|\psi_{1s,1s}^{SA}\rangle = |\psi_{1s}^g\rangle_1 |\psi_{1s}^g\rangle_2 |00\rangle \doteq \frac{1}{2} [\psi_{1s}(\mathbf{r}_{1A}) + \psi_{1s}(\mathbf{r}_{1B})] [\psi_{1s}(\mathbf{r}_{2A}) + \psi_{1s}(\mathbf{r}_{2B})] |00\rangle. \quad (13.89)$$

Just as we found for helium, there is no possible way to make a spatial state that is antisymmetric with respect to electron exchange when both electrons are in the $|\psi_{1s}^g\rangle$ one-electron ground state, so $|\psi_{1s,1s}^{SA}\rangle$ is the only possible state in the ground state of the molecule. We conclude that the ground state of the hydrogen molecule is a spin singlet state.

We can gain more insight into the molecular ground state by looking at the state $|\psi_{1s,1s}^{SA}\rangle$ more closely. If we expand Eq. (13.89), we obtain

$$\begin{aligned} |\psi_{1s,1s}^{SA}\rangle &\doteq \frac{1}{2} [\psi_{1s}(\mathbf{r}_{1A})\psi_{1s}(\mathbf{r}_{2A}) + \psi_{1s}(\mathbf{r}_{1B})\psi_{1s}(\mathbf{r}_{2B}) \\ &\quad + \psi_{1s}(\mathbf{r}_{1A})\psi_{1s}(\mathbf{r}_{2B}) + \psi_{1s}(\mathbf{r}_{2A})\psi_{1s}(\mathbf{r}_{1B})] |00\rangle. \end{aligned} \quad (13.90)$$

We can divide this into two terms, labeled “covalent” and “ionic”

$$\begin{aligned} |\psi_{cov}^S\rangle &\doteq \frac{1}{2}[\psi_{1s}(\mathbf{r}_{1A})\psi_{1s}(\mathbf{r}_{2B}) + \psi_{1s}(\mathbf{r}_{2A})\psi_{1s}(\mathbf{r}_{1B})] \\ |\psi_{ion}^S\rangle &\doteq \frac{1}{2}[\psi_{1s}(\mathbf{r}_{1A})\psi_{1s}(\mathbf{r}_{2A}) + \psi_{1s}(\mathbf{r}_{1B})\psi_{1s}(\mathbf{r}_{2B})], \end{aligned} \quad (13.91)$$

so that

$$|\psi_{1s,1s}^{SA}\rangle = (|\psi_{cov}^S\rangle + |\psi_{ion}^S\rangle)|00\rangle. \quad (13.92)$$

The state $|\psi_{cov}^S\rangle$ corresponds to the situation with one electron associated with each nucleus, whereas the state $|\psi_{ion}^S\rangle$ corresponds to the situation with both electrons associated with one nucleus. When the nuclei are well separated, $|\psi_{cov}^S\rangle$ corresponds to two isolated hydrogen atoms and $|\psi_{ion}^S\rangle$ corresponds to a proton and a negative hydrogen ion, which has an energy larger than the two isolated hydrogen atoms. Hence, we expect that $|\psi_{cov}^S\rangle$ would be a better guess for the ground state of the molecule. The state $|\psi_{cov}^S\rangle$ represents **covalent bonding** and the state $|\psi_{ion}^S\rangle$ represents **ionic bonding**.

For the covalent bond, we can also form an antisymmetric state

$$|\psi_{cov}^A\rangle \doteq \frac{1}{2}[\psi_{1s}(\mathbf{r}_{1A})\psi_{1s}(\mathbf{r}_{2B}) - \psi_{1s}(\mathbf{r}_{2A})\psi_{1s}(\mathbf{r}_{1B})], \quad (13.93)$$

which must be associated with the symmetric spin-triplet state:

$$|\psi_{cov}^{AS}\rangle = |\psi_{cov}^A\rangle|1M\rangle. \quad (13.94)$$

If we use the two states $|\psi_{cov}^{SA}\rangle = |\psi_{cov}^S\rangle|00\rangle$ and $|\psi_{cov}^{AS}\rangle = |\psi_{cov}^A\rangle|1M\rangle$ to find the energy expectation values, then we are using the **valence bond method**. The results of this calculation for the case with both electrons in the 1s atomic states are shown in Fig. 13.16. The results are qualitatively similar to

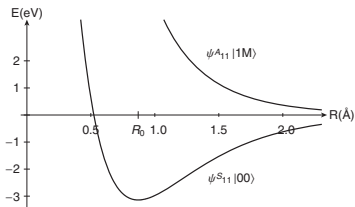


FIGURE 13.16 Energies of the bonding and antibonding orbitals of the hydrogen molecule obtained with the valence bond method, with the zero of energy referenced to the dissociation limit.

Fig. 13.14 in that there is a bonding orbital and antibonding orbital. This is to be expected because the state $|\psi_{cov}^S\rangle$ is a *gerade* state and the state $|\psi_{cov}^A\rangle$ is an *ungerade* state. But now for the H_2 molecule, these states are also linked to the exchange symmetry and hence the spin. Following the argument for the hydrogen ion, we conclude that the symmetric spatial state (singlet spin state) has a lower energy than the antisymmetric spatial state (triplet spin state) because of the increased electron-proton Coulomb attraction in the *gerade* state. Note that this ordering of the singlet and triplet states is opposite the case for the excited states of helium. In that case, the increased overlap of the electrons in the symmetric spatial state led to an increased Coulomb repulsion of the two electrons and a higher energy for the spin singlet state.

SUMMARY

For a proper quantum mechanical description of multiple-particle systems, we must account for the indistinguishability of fundamental particles. The symmetrization postulate requires that the quantum state vector of a system of identical particles be either symmetric or antisymmetric with respect to exchange of any pair of identical particles within the system. Nature dictates that integer spin particles—bosons—have symmetric states, while half-integer spin particles—fermions—have antisymmetric states. The symmetrization postulate applies to the complete quantum state vector, including both the spin and space parts of the system. As a consequence of the symmetrization postulate, some states are not allowed. The best known manifestation of this is the Pauli exclusion principle, which limits the number of electrons in given atomic levels and leads to the structure of the periodic table.

PROBLEMS

- 13.1 Show that the eigenvalues of the exchange operator P_{12} are ± 1 .
- 13.2 For a system of two identical spin-0 bosons, the total spin must be zero and the only possible system spin state is $|SM\rangle = |00\rangle$. Express this state in the uncoupled basis and show that it is symmetric with respect to exchange of the two particles.
- 13.3 Consider a system of two identical spin-1 particles. Find the spin states for this system that are symmetric or antisymmetric with respect to exchange of the two particles.
- 13.4 Specify the exchange symmetry of the following wave functions:

$$\begin{aligned}\psi_a(x_1, x_2) &= \frac{1}{(x_1 + x_2)} \\ \psi_b(x_1, x_2) &= \frac{a(x_1 - x_2)}{(x_1 - x_2)^2 + b} \\ \psi_c(x_1, x_2) &= \frac{a(x_1 - 3x_2)}{(x_1 + x_2)^2 + b} \\ \psi_d(x_1, x_2, x_3) &= \frac{x_1 x_2 x_3}{x_1^2 + x_2^2 + x_3^2 + b}.\end{aligned}$$

- 13.5 Use your favorite software tool to plot the two-particle probability density for two non-interacting particles in a one-dimensional harmonic oscillator potential for the case where one of the particles is in the single-particle ground state and the other is in the single-particle first excited state. Do this for (a) distinguishable particles (of the same mass), (b) identical spin-0

bosons, and (c) identical spin-1/2 fermions in a spin triplet state. In each case, write the system wave function and discuss the important features of your plots.

- 13.6** Consider two noninteracting particles of mass m in an infinite square well. For the case with one particle in the single-particle state $|n\rangle$ and the other in the state $|k\rangle$ ($n \neq k$), calculate the expectation value of the squared interparticle spacing $\langle (x_1 - x_2)^2 \rangle$, assuming (a) the particles are distinguishable, (b) the particles are identical spin-0 bosons, and (c) the particles are identical spin-1/2 fermions in a spin triplet state. Use bra-ket notation as far as you can, but you will have to do some integrals. Verify the results in Eq. (13.39).
- 13.7** Consider two noninteracting particles of mass m in the harmonic oscillator potential well. For the case with one particle in the single-particle state $|n\rangle$ and the other in state $|k\rangle$ ($n \neq k$), calculate the expectation value of the squared interparticle spacing $\langle (x_1 - x_2)^2 \rangle$, assuming (a) the particles are distinguishable, (b) the particles are identical spin-0 bosons, and (c) the particles are identical spin-1/2 fermions in a spin triplet state. Use bra-ket notation as far as you can, but you will have to do some integrals.
- 13.8** Calculate the one-dimensional particle separation probability density $\mathcal{P}(x_1 - x_2)$ for a system of two identical particles in an infinite square well with one particle in the single-particle ground state $|1\rangle \doteq \varphi_1(x)$ and the other in the state $|2\rangle \doteq \varphi_2(x)$. Do this for the three cases of (a) distinguishable particles (of the same mass), (b) identical particles in a symmetric spatial state, and (c) identical particles in an antisymmetric spatial state. Reproduce Fig. 13.4.
- 13.9** Calculate the one-particle probability density $\mathcal{P}(x_1)$ by integrating the two-particle probability density $\mathcal{P}(x_1, x_2)$ over the position x_2 of particle 2 (i.e., projecting the two-particle probability density onto the x_1 axis). Do this for the three cases of (a) distinguishable particles (of the same mass), (b) identical particles in a symmetric spatial state, and (c) identical particles in an antisymmetric spatial state. Demonstrate that measuring the position of one particle independent of the location of the other particle is the same for all three cases.
- 13.10** Show that Eq. (13.51) follows from Eq. (13.50).
- 13.11** Consider two indistinguishable, uncharged spin-1/2 fermions in the one-dimensional harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$. The two particles interact with each other through a perturbing potential $H' = \frac{1}{2}\alpha(x_1 - x_2)^2$, where the positive constant α is considered small ($\alpha \ll m\omega^2$).
- For the unperturbed two-particle system, find the energy eigenvalues and eigenstates of the ground state and the first excited state (you need not determine the spatial wave functions, bra-ket notation is sufficient). Specify and discuss the degeneracy of each level.
 - Discuss qualitatively how the energies in (a) are perturbed by the interaction of the particles. Draw an energy level diagram showing the unperturbed and perturbed energy levels.
- 13.12** Show that the sign of the exchange contribution K_{12} is negative for the spin-triplet state in the first excited state of a system of two identical spin-1/2 particles [see Eq. (13.57)].
- 13.13** Consider the first excited state of helium where one electron is in the $n = 1$ hydrogenic state and the other electron is in the $n = 2$ hydrogenic state.
- Using term or spectroscopic notation, list all the allowed states of this system.
 - How many total states are there?
 - What is the energy of this level, ignoring the interactions of the electrons with each other?
 - Describe qualitatively the shifts of this energy level that result from considering the interactions of the electrons with each other.

- 13.14** Find the first-order perturbed energy of the helium ground state by calculating the direct integral J in Eq. (13.70). Find the numerical value of your result (in eV) and confirm Eq. (13.73).
- 13.15** Find the first-order perturbed energies of the helium excited states $1s2s$ and $1s2p$ by calculating the direct and exchange integrals J , K in Eqs. (13.71) and (13.81). Find the numerical values of your results (in eV) and make a diagram similar to Fig. 13.9.
- 13.16** Show that the state of the hydrogen molecule that is antisymmetric with respect to electron exchange when both electrons are in the $|\psi_{1s}^{\uparrow}\rangle$ state is identically zero.
- 13.17** Consider two indistinguishable, noninteracting spin-1/2 fermions in a one-dimensional infinite square well potential of length L .
- What is the ground-state energy of the two-particle system?
 - What is the ground-state wave function?
 - What is the first excited state energy of the two-particle system?
 - What are the wave functions of the first excited state?
 - What is the degeneracy of the first excited state?
 - Discuss qualitatively how the excited-state energies change if we consider the particles to be interacting through the Coulomb potential.

RESOURCES

Further Reading

The work on Bose-Einstein Condensation that was awarded the 2001 Nobel Prize in Physics is described at:

nobelprize.org/nobel_prizes/physics/laureates/2001/

Further details on molecular energy calculations are presented in

B. H. Bransden and C. J. Joachain, *Physics of Atoms and Molecules*, 2nd ed., Harlow, England: Prentice Hall, 2003.

Dr. Seuss's take on indistinguishability can be found in

Dr. Seuss, *The Sneetches and Other Stories*, New York: Random House, 1961.